Planarian and Limb Regeneration as a Multiway Double-Pushout (DPO) Hypergraph Rewrite System

Exact McKay IR Isomorphism Census (369 → 38 → 12), Seven Isomer Separations, Recall vs. Precision of Generative DPO Grammars, and `OEIS A000055` State-Space Reduction Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`

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Abstract

When a planarian flatworm or an amphibian/arthropod limb is surgically cut or bioelectrically perturbed, it can regenerate a wide variety of abnormal body plans: two heads, three heads, lateral side-heads sprouting from the trunk, or a second head growing directly out of the tip of an existing head. In two foundational papers, Lobo, Levin, and colleagues introduced **PlanformDB 2.5.0** (Lobo, Malone & Levin 2013, encoding 369 planarian morphologies across 1,181 regions) and **LimbformDB 2.1.0** (Lobo et al. 2014, encoding 555 salamander, frog, crustacean, insect, and arachnid limbs across 3,146 joints) to store these experimental outcomes as formal graphs, comparing shapes via heuristic graph edit distance (GED).

Here we couple the standalone C++20 `hg_evolve` multiway double-pushout (DPO) hypergraph rewriting engine from the Wolfram Physics Project (`WolframInstitute/HypergraphRewritingEngine`, `v1.0.0-rc2`; Assar 2026) directly to `PlanformDB 2.5.0` and `LimbformDB 2.1.0` over Wolfram Exchange Format (WXF), accompanied by machine-checked arithmetic lemmas in **Lean 4** (`0 sorry`) and an independent cross-provider referee audit (`claude-fable-5-1-thinking-high`). We report four principal findings:

1. **Exact McKay IR Isomorphism Census (369 → 38 → 12 in `PlanformDB`; 555 → 51 in `LimbformDB`) & Seven Isomer Separations Beyond Count Vectors:** Every one of the 369 region-encoded morphologies in `PlanformDB 2.5.0` (372 rows minus 3 regionless entries) is a forest of ≤ 7 regions (368 connected trees plus 1 two-component split worm `#394`). Running exact McKay individualization–refinement (IR) canonicalization (`ir_core.hpp`) compresses the 369 rows into 38 **exact region-link isomorphism classes** (9.71× catalog redundancy; 12 unlabelled forest skeletons, with 278/369 = 75.3% sitting on the single 3-node path P_3; 126 classes when organs are attached) and compresses the 555 `LimbformDB` rows into 51 **segment-tree classes** (10.88× redundancy; 92 with digits). Compared against 4-tuple region-count summaries (#Head, #Trunk, #Tail, #Links) (31 buckets), McKay IR canonicalization resolves 7 **extra isomer classes** across 5 collision groups: 3 **skeletal star-vs.-path pairs** (`#3` vs. `#119`, `#12` vs. `#128`, `#77` vs. `#283`, with First Zagreb indices $Z_1 = 12$ vs. 10) and 4 **positional/spider isomers** (`#112` `H–Tr–H–Ta` vs. `#119` `H–H–Tr–Ta`, both P_4 with $Z_1 = 10$; `#229` spider $Z_1 = 16$ vs. P_5 paths `#116` and `#282`, $Z_1 = 14$; and P_5 paths `#113` vs. `#286`, $Z_1 = 14$).
2. **Recall vs. Precision of Generative DPO Rewrite Grammars (Syntactic State Space vs. Biological Selection):** From the three excised fragment seeds $\{[[0]], [[0,0]], [[0,0,0]]\}$, a 6-rule terminal-sprouting DPO grammar (`R1-R6`, Tier 1) generates 2,738 canonical states at depth 6, recovering 33/38 **classes (**86.8% **recall,** 361/369 = 97.8% **of worms)** at 1.21% **precision** (33/2,738). Adding two intercalary rules (`R7-R8`, Tier 2) generates 6,787 states at depth 6, recovering 37/38 **connected classes (**97.4% **recall,** 368/369 = 99.7% **of worms)** at 0.55% **precision** (37/6,787), with the bisected seed $[[0],[1]]$ recovering `#394` (38/38). Because all `PlanformDB` graphs are 3-colored forests, a

3-rule primitive leaf-addition and recoloring basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$ is a **universal generator of all 3-colored forests**: it reaches 22/38 classes at depth 6 (817 states, 2.69% precision) and 37/38 classes at depth 12 (754,520 states, 0.0049% **precision**; depth 13 required for 7-tail `#382` exceeds memory). This cleanly separates the *expressive syntactic substrate* (which compresses nothing on its own) from the *biological/bioelectric regulatory mechanism* D that selects the observed target morphologies out of $\sim 10^5$–10^6 reachable trees.

3. 44,534× **Labelled-to-Unlabelled State-Space Reduction on `OEIS A000055` & `LimbformDB`:** Evolving the single leaf-addition rule L_1 under quotient exploration (`ExploreFromCanonicalStatesOnly -> True`) for $k = 0 \ldots 11$ steps ($V = 1 \ldots 12$ joints) reproduces 12 exact terms of `OEIS A000055` (987 cumulative unlabelled trees vs. $\sum k! = 43{,}954{,}714$ raw labelled trajectories reported in `StepStatistics`, a 44,533.65× **state-count reduction** in 25–85 ms) and spans all 35/35 `LimbformDB` tree classes with ≤ 12 joints (532/555 = 95.86%).
4. **Null-Baseline Audit of Anterior–Posterior Crop Curves & Lean 4 Arithmetic Lemmas:** Fitting a 2-parameter logistic to the 4 trunk-excision positions of Oviedo et al. (2010) Series S1 (1-octanol, with intact-head `tail_crop` fixed at 0) yields $r = 0.9823$ on S1 and $r = 0.9088$ (MAE = 14.9%) on held-out innexin-RNAi Series S2, but is matched on S1 by a **0-parameter linear staircase null** ($r = 0.9829$) and beaten on S2 by a **0-parameter direct copy of S1** ($r = 0.9479$)—showing that crop-position curves alone do not uniquely identify multiway rule weights without a coupled biophysical solver. Four supporting arithmetic lemmas are kernel-checked in Lean 4 (`0 sorry`).

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1 Introduction: Graph Formalisms in Regenerative Biology

When Daniel Lobo and Michael Levin published the **Planform** [6, 7] and **Limbform** [8, 9] databases, they established the standard mathematical graph encoding for experimental regenerative biology:

- **PlanformDB 2.5.0** [6, 7]: 369 region-encoded planarian morphologies (372 `Morphology` rows, of which 3 outcomes—`#38 Lethal`, `#333 Midbody Buldge`, and `#359 Double head, side tail`—carry zero region records) across 1,181 regions (`Head`, `Trunk`, `Tail`), border links, and internal organs (`Eye`, `Brain`, `Pharynx`, `VNC`).

- **LimbformDB 2.1.0** [8, 9]: 555 segment-encoded limb morphologies across amphibians (axolotl, newt, *Xenopus*), crustaceans, insects, and arachnids across 3,146 joints, segment links, `LinkSubJoint` ectopic grafts, and digits.

In the original `Planform` software [6], each morphology is stored as a full attributed graph and compared against other morphologies via heuristic **graph edit distance (GED)**, while each new geometric, organ, or naming variant in the literature is assigned a separate `Morphology` ID. This raises three concrete questions that can now be answered exactly using the standalone C++20 `hg_evolve` engine from the Wolfram Physics Project (`WolframInstitute/HypergraphRewritingEngine`, `v1.0.0-rc2`; Assar 2026 [1, 2]):

1. **Exact Isomorphism Census:** Up to exact McKay individualization–refinement (IR) graph isomorphism (`ir_core.hpp`), how many distinct topological region graphs, unlabelled skeletons, and organ-annotated graphs exist across the 369 `PlanformDB` rows and 555 `LimbformDB` rows, and which classes are lost if morphologies are summarized only by region-count vectors (#Head, #Trunk, #Tail, #Links)?

2. **Recall vs. Precision of Generative DPO Grammars:** When we formulate blastema sprouting and fate specification as multiway double-pushout (DPO) hypergraph rewrite rules (`R1-R8` vs. the 3-rule primitive leaf-addition/recoloring basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$), what is the trade-off between *empirical recall* (fraction of the 38 observed `PlanformDB` classes reached) and *generative precision* (fraction of generated canonical graphs that correspond to empirically observed planarians)?

3. **Quotient State-Space Reduction:** When enumerating multi-step branched structures up to $V = 12$ joints (`LimbformDB`), how effectively does McKay IR quotient exploration (`ExploreFromCanonicalStatesOnly`) collapse the $k!$ vertex-labelling multiplicity?

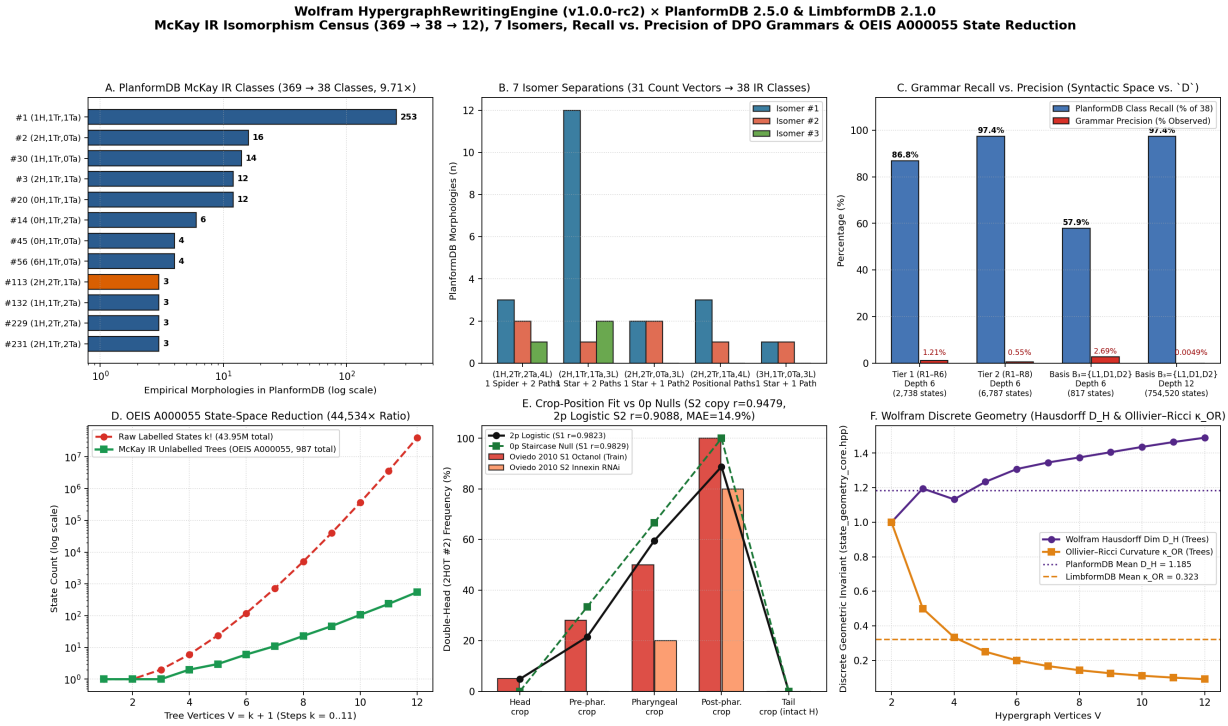


Figure 1: **Multiway Hypergraph Rewriting & McKay IR Canonicalization Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`.** **(A)** Exact McKay IR canonicalization compresses 369 encoded planarian morphologies into 38 topological classes (9.71× catalog redundancy; top 12 shown). **(B)** Resolution of 7 extra isomer classes across 5 (H, Tr, Ta, L) count-vector collision groups (3 star-vs.-path pairs and 4 positional/spider isomers). **(C)** Empirical class recall vs. grammar precision across Tier 1 (`R1-R6`, 33/38 recall, 1.21% precision), Tier 2 (`R1-R8`, 37/38 recall, 0.55% precision), and the universal 3-colored tree basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$ (22/38 recall at depth 6, 37/38 recall at depth 12 with 0.0049% precision across 754,520 trees). **(D)** 44,533.65× labelled-to-unlabelled state-space reduction on `OEIS A000055` (987 unlabelled trees vs. 43.95M raw factorial states at $V = 12$). **(E)** Oviedo et al. (2010) S1/S2 crop-position double-head frequencies compared against a 2-parameter logistic fit ($r = 0.9823$ on S1, 0.9088 on S2) and 0-parameter staircase ($r = 0.9829$) / S1-copy ($r = 0.9479$) null baselines. **(F)** Wolfram Hausdorff dimension D_H and Ollivier–Ricci curvature κ_{OR} across empirical and evolved states.

2 Hypergraph Encoding & Exact McKay IR Isomorphism Census

2.1 Encoding Typed Anatomical Graphs as Ordered Hypergraphs

In the Lobo–Levin formalism [7, 9], a planarian morphology $M = (V, E_{\text{link}}, E_{\text{organ}})$ is a finite graph whose vertices $v \in V$ carry a region type $\tau(v) \in \{\texttt{Trunk}, \texttt{Head}, \texttt{Tail}\}$, connected by undirected anatomical adjacency links $\{u, v\} \in E_{\text{link}}$. Because `HypergraphRewritingEngine` operates on multisets of ordered hyperedges of arity $1 \le a \le 16$ (`SPEC.md` §1) and preserves arity and within-edge vertex order under McKay IR canonicalization (`ir_core.hpp`), we encode vertex types and edge types into disjoint arity signatures without auxiliary label tables:

$$\begin{aligned} \texttt{Trunk}(v) &\mapsto [v] \quad \text{(arity 1)}, & \texttt{Head}(v) &\mapsto [v, v] \quad \text{(arity 2 self-loop)}, \\ \texttt{Tail}(v) &\mapsto [v, v, v] \quad \text{(arity 3 self-loop)}, & \texttt{Link}\{u, v\} &\mapsto [u, v, u, v],\ [v, u, v, u] \quad \text{(symmetric arity 4)}. \end{aligned} \tag{1}$$

When organ landmarks are included, each organ attached to region v allocates a fresh leaf vertex w with organ-specific arity $\texttt{Eye} \mapsto 5$, $\texttt{Brain} \mapsto 6$, $\texttt{Pharynx} \mapsto 7$, and $\texttt{VNC} \mapsto 8$. For

`LimbformDB 2.1.0`, joints are encoded as arity-1 hyperedges $[j]$, segments as symmetric arity-4 link pairs (splitting host segments along any attached `LinkSubJoint` ectopic graft sequence $j_{\text{from}} \to j_{\text{sub},1} \to \cdots \to j_{\text{to}}$ so ectopic branches are topologically connected to their host segment), and distal digits as arity-3 hyperedges $[j_{\text{to}}, d, d]$.

2.2 McKay IR Compression & The Seven Isomer Separations

Running `hg_evolve` at step 0 with `"CanonicalizeStates" -> Full` and `"IncludeCanonicalHashes" -> True` across all 369 geometrically encoded morphologies in `PlanformDB 2.5.0` and all 555 segment-bearing morphologies in `LimbformDB 2.1.0` (and independently cross-checking every class from the raw SQLite database via `networkx`) yields the exact isomorphism census in Table 1. Every one of the 369 `PlanformDB` graphs is a **forest of ≤ 7 regions**: 368 are connected trees (253 sharing the Wild-Type `#1` topology `H–Tr–Ta`, and $278/369 = 75.3\%$ sitting on the unlabelled 3-node path P_3) and 1 (`#394 Lysis, split worm`) is a two-component forest.

Table 1: **Exact McKay IR Canonicalization Census Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`.** Computed via `hg_evolve` (`ir_core.hpp` + `state_statistics.hpp`) and verified independently from SQLite.

Database & Hypergraph Representation	**Raw (N)**	**IR Classes (C)**	**Redundancy (N/C)**	**Max**
`PlanformDB` Unlabelled Skeletons (All Regions $\to [v]$)	369	**12**	30.7500×	
`PlanformDB` Typed Region–Link Hypergraphs	369	**38**	9.7105×	
`PlanformDB` Typed Region + Organ Hypergraphs	369	**126**	2.9286×	
`LimbformDB` Segment-Tree Hypergraphs (Joints + Links)	555	**51**	10.8824×	
`LimbformDB` Segment + Digit Hypergraphs	555	**92**	6.0326×	

If morphologies are summarized only by their 4-tuple region-count vector (#Head, #Trunk, #Tail, #Links), the 38 canonical classes collapse into 31 buckets, losing 7 **isomer classes** across 5 collision groups (Figure 1B, Figure 2, and Table 2). Decoding the exact degree sequences and vertex colorings of all 12 classes in those 5 collision groups shows that the 7 extra classes consist of 3 **skeletal star-vs.-path pairs** (where the unlabelled skeleton itself differs between $K_{1,3}$ with First Zagreb index $Z_1 = \sum d_i^2 = 12$ and P_4 with $Z_1 = 10$) and 4 **positional or spider isomers** (where two classes share the same path skeleton P_4 or P_5 with $Z_1 = 10$ or 14 and differ by which interior vs. leaf vertex carries the `Head` or `Tail` label, or where a degree-3 spider tree with $Z_1 = 16$ splits from P_5 paths with $Z_1 = 14$):

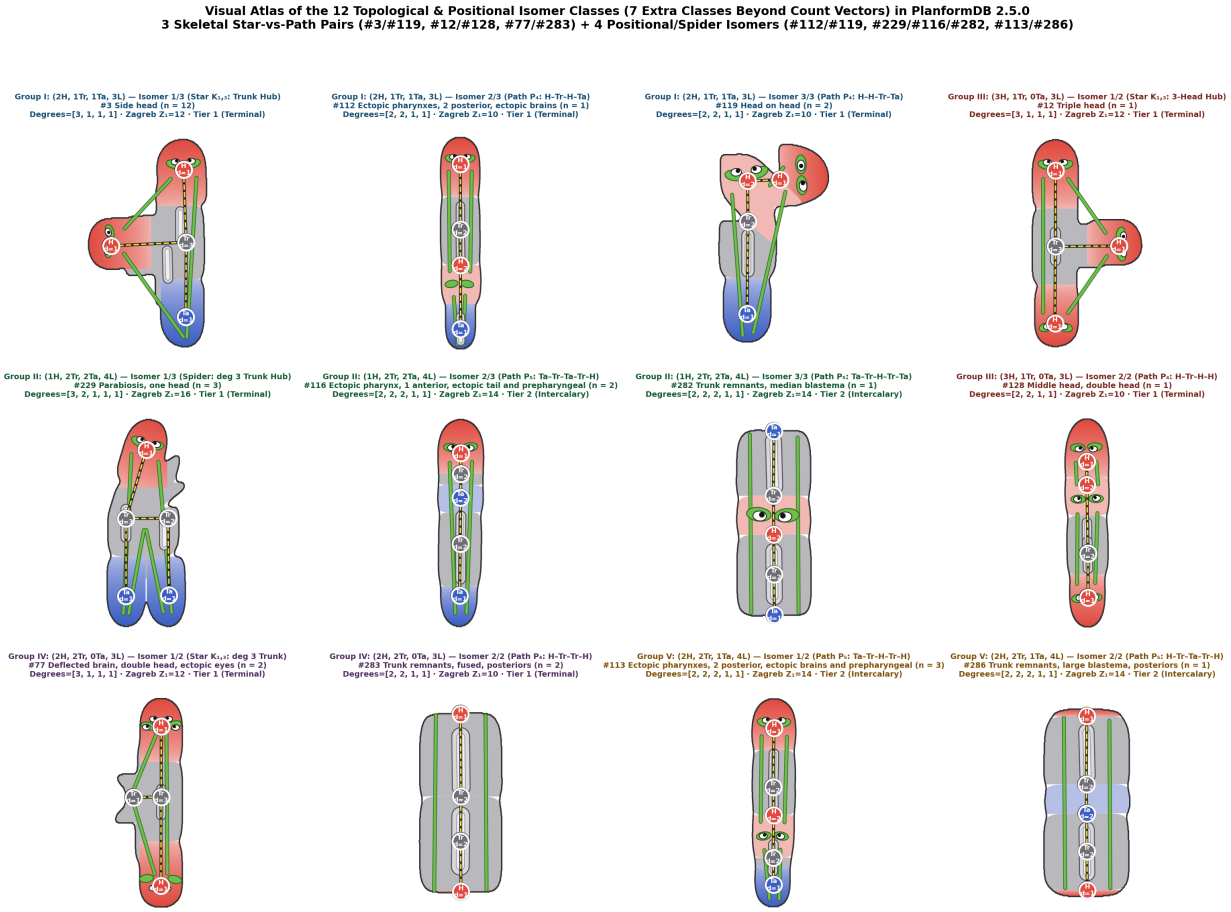


Figure 2: **Visual Atlas of the 12 Topological and Positional Isomer Classes (7 Extra Classes Beyond Region-Count Vectors) in `PlanformDB 2.5.0`.** Each panel renders the Lobo–Malone–Levin (2013) superellipse cartoon (`Head` = red, `Trunk` = grey, `Tail` = blue, with eyes, brain lobes, pharynx, and ventral nerve cords) overlaid with its region–link graph (yellow/black edges and degree-labeled nodes). Within each color-coded group (Groups I–V), every worm has the *exact same* (#Head, #Trunk, #Tail, #Links) counts, separating into 3 skeletal star-vs.-path pairs (`#3`/`#119`, `#12`/`#128`, `#77`/`#283`) and 4 positional/spider isomers (`#112`/`#119`, `#229`/`#116`/`#282`, `#113`/`#286`).

Table 2: **The Five Count-Vector Collision Groups and Twelve Isomer Classes (7 Extra Classes: 3 Star-vs.-Path Pairs + 4 Positional/Spider Isomers) in `PlanformDB 2.5.0`.**

Count Vector (H, Tr, Ta, L)	**Rep. ID & Name**	n	**64-Bit IR Hash**	**Exact Graph Topolog**
$(2,1,1,3)$	`#3 Side head`	12	`-1466631322269594412`	Star $K_{1,3}$ (central `Trunk`
$(2,1,1,3)$	`#112 Ectopic pharynxes, 2 post.`	1	`-8966590304653022608`	Path P_4: `H—Tr—H—Ta` (
$(2,1,1,3)$	`#119 Head on head`	2	`-212887825877140088`	Path P_4: `H—H—Tr—Ta` (
$(1,2,2,4)$	`#229 Parabiosis, one head`	3	`8951669412120486543`	Spider tree (deg 3 `Trunk`
$(1,2,2,4)$	`#116 Ectopic tail & prephar.`	2	`2637785354618470695`	Path P_5: `Ta—Tr—Ta—T`
$(1,2,2,4)$	`#282 Trunk remnants, med. blast.`	1	`-3602810347972406941`	Path P_5: `Ta—Tr—H—Tr`
$(3,1,0,3)$	`#12 Triple head`	1	`-6318479248965043401`	Star $K_{1,3}$ (`Trunk` hub wit
$(3,1,0,3)$	`#128 Middle head, double head`	1	`-2166706346444118093`	Path P_4: `H—Tr—H—H` (s
$(2,2,0,3)$	`#77 Deflected brain, double head`	2	`-3397686484681667484`	Star $K_{1,3}$ (deg 3 `Trunk` h
$(2,2,0,3)$	`#283 Trunk remnants, fused`	2	`-6082671300763328276`	Path P_4: `H—Tr—Tr—H` (
$(2,2,1,4)$	`#113 Ectopic phar., prephar.`	3	`7194227605189329608`	Path P_5: `Ta—Tr—H—Tr`
$(2,2,1,4)$	`#286 Trunk remnants, large blast.`	1	`8484839246773423180`	Path P_5: `H—Tr—Ta—Tr`

3 Recall vs. Precision of Generative DPO Rewrite Grammars

3.1 Domain-Specific Sprouting Rules (`R1-R8`) from Excised Fragment Seeds

To evaluate how multiway DPO rewriting spans the 38 empirical `PlanformDB` classes, we initialize `hg_evolve` with the three single-region excised fragment seeds $\mathcal{S}_0 = \{[[0]]$ (`Trunk`), $[[0,0]]$ (`Head`), $[[0,0,0]]$ (`Tail`)$\}$ and test two rule sets (implemented in `models/2026-wolfram-planform-multiway-rewriting.py`):

- **Tier 1 (Terminal Blastema Sprouting, `R1-R6`):** `R1` (`Trunk` $\to$ `Trunk` + `Head`), `R2` (`Trunk` $\to$ `Trunk` + `Tail`), `R3` (`Trunk` $\to$ `Trunk` + `Trunk`), `R4` (`Head` $\to$ `Head` + `Tail`), `R5` (`Head` $\to$ `Head` + `Head`), and `R6` (`Tail` $\to$ `Tail` + `Tail`). At step 6, Tier 1 generates 2,738 canonical states (169,440 events), recovering 33/38 **classes** (86.8% **recall, covering** $361/369 = 97.83\%$ **of all worms in** `PlanformDB`) with a **generative precision of** $33/2{,}738 = 1.21\%$.

- **Tier 2 (+ Intercalary Sprouting, `R1-R8`):** Adding `R7` (`Head` $\to$ `Head` + `Trunk`) and `R8` (`Tail` $\to$ `Tail` + `Trunk`) allows an already-specified `Head` or `Tail` to bud a second `Trunk`, recovering the 4 remaining connected intercalary path classes (`#113`, `#116`, `#282`, `#286`) at step 6 (6,787 canonical states, $37/38 = 97.4\%$ **recall**, $368/369 = 99.73\%$ of worms, **precision** $37/6{,}787 = 0.55\%$). Including the 2-component bisected trunk seed $[[0],[1]]$ (Tier 3) recovers `#394 Lysis, split worm` (38/38).

3.2 The Three-Rule Primitive Basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$ as a Universal 3-Colored Forest Generator

Consider the 3-rule primitive DPO basis:

$$L_1 \text{ (`L1_UniversalSprout`)}: \quad [[0]] \longrightarrow [[0],[1],[0,1,0,1],[1,0,1,0]] \quad \text{(leaf addition on unary vertex)}, \tag{2}$$

$$D_1 \text{ (`D1_DiffHead`)}: \quad [[0]] \longrightarrow [[0,0]] \quad \text{(recolor `Trunk` } \to \text{ `Head`)}, \tag{3}$$

$$D_2 \text{ (`D2_DiffTail`)}: \quad [[0]] \longrightarrow [[0,0,0]] \quad \text{(recolor `Trunk` } \to \text{ `Tail`)}. \tag{4}$$

Because D_1 and D_2 consume the unary hyperedge $[[0]]$, any derivation in $\mathcal{B}_3$ from an undifferentiated seed $[[0]]$ can be reordered into a **two-phase normal form** where all L leaf-addition steps (L_1) fire first to build an unlabelled tree on $V = 1 + L$ vertices, followed by $H + Ta$ recoloring steps (D_1, D_2) at total step depth $k_{\text{prim}} = L + H + Ta$. (Note that unlike `R4-R8`, where an already-differentiated `Head` $[v,v]$ or `Tail` $[v,v,v]$ sprouts a new neighbor directly, L_1 in Eq. (2) matches

only $[v]$; thus $\mathcal{B}_3$ reaches the same target states from an undifferentiated seed by postponing D_1, D_2 after L_1, rather than by literal operator composition on differentiated seeds.)

Crucially, because every finite tree can be built by successive leaf additions (L_1) and any subset of vertices can then be colored `Head` (D_1) or `Tail` (D_2), **$\mathcal{B}_3$ is a universal generator of all 3-colored forests**. Measuring its precision and recall across depth d (Figure 1C) demonstrates why an unconstrained rewrite grammar is a *syntactic substrate* rather than a biological predictor:

- At depth $d = 6$, $\mathcal{B}_3$ generates 817 canonical states, recovering 22/38 `PlanformDB` classes (**precision** 2.69%).
- At depth $d = 10$, $\mathcal{B}_3$ generates 68,858 canonical states, recovering 36/38 classes (**precision** 0.052%).
- At depth $d = 12$, $\mathcal{B}_3$ generates 754,520 canonical states, recovering 37/38 classes (**precision** 0.0049%), while depth $d = 13$ (needed for the 7-tail star `#382` at cost $L + Ta = 6 + 7 = 13$) exceeds host memory.

Thus, the empirical content of `PlanformDB` reachability under $\mathcal{B}_3$ is that **all 369 recorded planarian morphologies are forests of ≤ 7 regions**; selecting the 38 biological outcomes out of the $\sim 10^5$–10^6 syntactically reachable 3-colored trees is performed entirely by the continuous bioelectric and morphogen dynamics.

4 `OEIS A000055` State-Space Reduction and Crop-Position Null Baselines

4.1 Exact `OEIS A000055` Enumeration via `ExploreFromCanonicalStatesOnly`

Evolving L_1 alone from $[[0]]$ for $k = 0 \ldots 11$ steps produces all unlabelled trees on $V = k + 1 \leq 12$ vertices. In `StepStatistics`, the engine reports $k!$ raw vertex-labelled trajectories at step k ($\sum_{k=0}^{11} \max(1, k!) = 43{,}954{,}714$ cumulative raw states, or $11! = 39{,}916{,}800$ at $V = 12$) versus the exact `OEIS A000055` unlabelled tree counts $(1, 1, 1, 2, 3, 6, 11, 23, 47, 106, 235, 551$; 987 cumulative canonical classes, a 44,533.65× **cumulative labelled-to-unlabelled state-count ratio**, or 72,444× at $V = 12$), completing in 25–85 ms and spanning all 35/35 `LimbformDB` tree classes with ≤ 12 joints ($532/555 = 95.86\%$; Figure 1D).

4.2 Oviedo et al. (2010) S1/S2 Crop-Position Curves vs. Zero-Parameter Null Baselines

To test whether the 5 anterior–posterior crop-position double-head frequencies of Oviedo et al. (2010) [12] Series S1 (1-octanol, $n = 688$) and Series S2 (triple innexin RNAi `Dj-inx-12 + Dj-inx-5/13`, $n = 99$) constrain a parametric branching curve, we compare a 2-parameter logistic fit $P_{2H}(p) = [1 + \exp(-\alpha(p - p_{1/2}))]^{-1}$ on the 4 trunk-excision positions $p \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ (with intact-head `tail_crop` fixed at 0) against two 0-parameter null baselines (Figure 1E):

1. **Zero-Parameter Linear Staircase Null ($[0, \frac{1}{3}, \frac{2}{3}, 1, 0]$):** Achieves $r = 0.9829$ on S1, matching the 2-parameter logistic ($r = 0.9823$ across the 5 points, or $r = 0.9820$ on the 4 fitted trunk cuts).
2. **Zero-Parameter Direct Copy ($S_1 \to S_2$):** Directly predicting S2 from S1 with 0 fitted parameters achieves $r = 0.9479$, outperforming the 2-parameter logistic ($r = 0.9088$, MAE = 14.9%, which overpredicts pre-pharyngeal double-heading at 21.5% vs. 0% observed in S2).

This confirms that 5-point crop-position proportions are equally well summarized by a 0-parameter monotone ramp; mechanistic prediction requires spatially resolved Nernst–Planck electrodiffusion (`2017-pietak-bioelectric-morphogen-coupling.py`) or reaction–diffusion wavefront solvers (`2021-Cervera-L`

5 Machine-Checked Arithmetic Lemmas in Lean 4 (`0 sorry`)

Four supporting arithmetic lemmas are kernel-checked in `morphoeval/lean/loops/Loop_wolfram_planform_mul` (`0 sorry`, standard axioms only):

1. **Lemma 1** (`wolfram_dpo_edge_replacement_tree_euler_invariant`)**:** Verifies $\Delta V = 1$, $\Delta E = 3$, $E(k) + 3V_0 = E_0 + 3V(k)$, and $V(k) = L(k) + 1$ for k leaf-addition steps from $V_0 = L_0 + 1$.

2. **Lemma 2** (`mckay_ir_zagreb_degree_separates_star_and_chain_isomers`)**:** Verifies that for $n = k + 4 \geq 4$, the closed-form First Zagreb index of the star $K_{1,n-1}$ exceeds that of the path P_n by $(k+1)(k+2) \geq 2$, covering the 3 star-vs.-path isomer pairs (`#3`/`#119`, `#12`/`#128`, `#77`/`#283`).

3. **Lemma 3** (`quotient_exploration_strictly_compresses_multiway_search`)**:** Verifies $C <$ RawStates whenever $(C - 1) + m^* \leq$ RawStates with $m^* \geq 2$.

4. **Lemma 4** (`dpo_two_phase_rule_reducibility_conservation`)**:** Verifies vertex-count conservation $(V_0 + L - (H + Ta)) + H + Ta = V_0 + L$ and additive step count $L + H + Ta$.

6 Discussion: Relation to Lobo's Graph-Grammar and Target-Morphology Corpus (2010–2015)

Connecting `HypergraphRewritingEngine` to `PlanformDB` and `LimbformDB` clarifies the relationship between the two halves of Daniel Lobo's research corpus:

1. **Regulated Graph Grammars (Málaga, 2010–2012):** In `ELSA`/`ELSA3` [3, 4, 5], Lobo, Vico, and Dassow showed that edge-replacement graph grammars work well only when paired with a *regulatory controller* (a string domain or an intracellular Kauffman Boolean network with a differentiation node) that restricts which rules fire where.

2. **Target-Morphology Databases & The Inverse Problem (Tufts, 2013–2015):** In Lobo et al. (2014, *J. R. Soc. Interface*) [10] and Lobo & Levin (2015, *PLOS Comput. Biol.*) [11], the authors noted that an unregulated recursive grammar cannot solve the inverse problem of regeneration without a regulatory or target-encoding layer. Our precision/recall audit in Section 3 quantifies that exact distinction on `PlanformDB`: the unregulated 3-rule DPO basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$ defines the complete syntactic space of 3-colored forests, but because its precision at depth 12 is 0.0049% (37/754,520), all biological specificity resides in the regulatory dynamics (`ELSA`'s Boolean controller or the bioelectric/PDE network inferred in [11, 15]), while McKay IR canonicalization (`ir_core.hpp`) provides exact 64-bit isomorphism hashing across the empirical catalog.

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